
GCVE-BCP-07 - Known Exploited Vulnerability (KEV) Assertion and No Known Exploitable Vulnerability (NKEV) Assessment Format

GCVE.eu



Contents

1	Known Exploited Vulnerability (KEV) Assertion and No Known Exploitable Vulnerability (NKEV) Assessment Format	1
2	Introduction	3
3	Known Exploited Vulnerability (KEV) Assertion Format	5
3.1	Usage Consideration	5
3.1.1	Relationship between KEV and NKEV	6
3.2	Format	7
3.2.1	Sample	7
3.2.1.1	Combined KEV Assertion	7
3.2.1.2	CISA KEV in BCP-07 Format	8
3.2.1.3	Withdrawn and Reasserted KEV Assertions	10
3.2.2	Field Description	11
3.2.2.1	uuid	11
3.2.2.2	vulnerability Object	11
3.2.2.2.1	vulnerability.vulnId	11
3.2.2.2.2	vulnerability.altId	11
3.2.2.3	status Object	12
3.2.2.3.1	status.exploited	12
3.2.2.3.2	status.status_reason	12
3.2.2.3.3	status.status_updated_at	12
3.2.2.4	characteristics Object	12
3.2.2.4.1	characteristics.remote_code_execution	12
3.2.2.4.2	characteristics.authentication_required	13
3.2.2.4.3	characteristics.local_access_required	13
3.2.2.4.4	characteristics.severity	13
3.2.2.5	timestamps Object	13
3.2.2.5.1	timestamps.first_seen_at	13
3.2.2.5.2	timestamps.asserted_at	13
3.2.2.5.3	timestamps.recorded_at	13

3.2.2.5.4	timestamps.last_seen_at	14
3.2.2.6	scope Object	14
3.2.2.6.1	scope.observation_regions	14
3.2.2.6.2	scope.victim_countries	14
3.2.2.6.3	scope.sector	14
3.2.2.6.4	scope.asset_exposure	14
3.2.2.6.5	scope.notes	15
3.2.2.7	evidence Array	15
3.2.2.7.1	evidence[].type	15
3.2.2.7.2	evidence[].signal	15
3.2.2.7.3	evidence[].confidence	15
3.2.2.7.4	evidence[].source	16
3.2.2.7.5	evidence[].details	17
3.2.2.7.6	evidence[].gcve	17
	gcve Object	17
3.2.2.8	gcve Object	17
3.2.2.8.1	gcve.origin_uuid	18
3.2.2.8.2	gcve.gna	18
3.2.2.8.3	gcve.object_uuid	18
4	No Known Exploitable Vulnerability (NKEV) Assessment Format	19
4.1	NKEV Assessment Object	19
4.1.1	NKEV Result Semantics	19
4.1.2	Sample NKEV Assessment	20
4.2	NKEV Field Description	22
4.2.1	uuid	22
4.2.2	assessment Object	22
4.2.2.1	assessment.type	22
4.2.2.2	assessment.result	23
4.2.2.3	assessment.assessed_at	23
4.2.2.4	assessment.knowledge_cutoff_at	23
4.2.3	assessment.product Object	23
4.2.3.1	assessment.product.vendor	23
4.2.3.2	assessment.product.name	24
4.2.3.3	assessment.product.version	24
4.2.3.4	assessment.product.identifiers	24
4.2.4	assessment.configurations Array	24
4.2.5	assessment.scope Object	25

4.2.6	assessment.knowledge_sources Array	25
4.2.7	assessment.findings Array	26
4.2.7.1	assessment.findings[].vulnerability	26
4.2.7.2	assessment.findings[].component	26
4.2.7.3	assessment.findings[].applicability	26
4.2.7.4	assessment.findings[].exploitability	26
4.2.7.5	assessment.findings[].justification	27
4.2.8	assessment.findings[].evidence and assessment.evidence	27
4.2.9	Referencing BCP-07 KEV Assertions from NKEV	28
4.2.10	Interoperability with VEX and Other Vulnerability Formats	29
4.3	GCVE KEV Directory and Source Index Format	29
4.3.1	Index Object	30
4.3.2	Index Field Description	30
4.3.2.1	kev Array	30
4.3.2.2	kev[].uuid	30
4.3.2.3	kev[].short_name	31
4.3.2.4	kev[].gcve_gna_id	31
4.3.2.5	kev[].url	31
4.3.2.6	kev[].automation_url	31
4.3.2.7	kev[].bcp_07_url	31
4.3.2.8	kev[].description	32
4.3.2.9	kev[].meta	32
4.3.2.9.1	kev[].meta.license	32
4.3.3	JSON Schema for the GCVE KEV Directory Array	32
4.3.4	JSON Schema - KEV Assertion	34
4.3.5	JSON Schema - NKEV Assessment	41
5	References	53
6	Acknowledgements	55
6.1	BCP-07 Coordinators	55
6.2	Contributions	55

1 Known Exploited Vulnerability (KEV) Assertion and No Known Exploitable Vulnerability (NKEV) Assessment Format



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2 Introduction

Known Exploited Vulnerabilities (KEVs) have become a critical signal for vulnerability prioritization, operational risk management, and policy-driven remediation. Governments, CSIRTs, and sectoral authorities increasingly rely on KEV lists to mandate patching, trigger incident response, or inform compliance decisions. However, existing KEV publications are largely list-based and opaque, often asserting exploitation without clearly expressing who made the claim, when exploitation was observed versus declared, what type of evidence supports it, where it was seen, or with what level of confidence. As KEV data is increasingly consumed by automated systems and cross-border information-sharing mechanisms, the absence of structured, contextual metadata limits interoperability, trust calibration, and analytical reuse.

This Best Current Practice defines a standardized KEV assertion format that preserves the intentionally simple and binary nature of KEV while adding minimal but essential context. Within the GCVE or other ecosystem, where vulnerabilities may be disclosed and referenced by multiple independent authorities, exploitation claims must be clearly distinguishable from vulnerability identifiers and treated as attributable statements rather than universal truths. The format enables multiple, potentially conflicting assertions to coexist, supports explicit attribution and confidence signaling, and facilitates interoperability with existing vulnerability, CSIRT, and policy ecosystems without turning KEV into full threat intelligence or requiring disclosure of sensitive evidence.

Version 3.0 additionally defines a **No Known Exploitable Vulnerability (NKEV) assessment format**. An NKEV assessment is a product-level, time-bounded technical assessment stating whether known exploitable vulnerabilities were identified for a specific product or product version, under a defined configuration and assessment scope, using explicitly identified vulnerability knowledge sources and evidence. NKEV assessments are deliberately modeled as a sibling object type to KEV assertions: KEV describes an exploitation assertion about a vulnerability, while NKEV describes the result of assessing a product against a body of vulnerability knowledge.

3 Known Exploited Vulnerability (KEV) Assertion Format

This format describes a **generic KEV (Known Exploited Vulnerability) assertion format**.

The goal is to express *who claims exploitation, when, based on what, where it was observed, and with which level of confidence*, without turning KEV into full threat intelligence. A KEV assertion is usually very binary and lacking some meta-information. The format adds some information which could better capture details about the exploitation. A majority of the fields are optional except *vulnerability*, *status* and *evidence*. `[]`.*source* which are recommended.

3.1 Usage Consideration

KEV and CVE (or GCVE) represent different layers in the model, and that distinction is intentional.

A vulnerability identifier (CVE/GCVE) is primarily about establishing the identity of a vulnerability something that the ecosystem can refer to consistently over time. KEV, on the other hand, is an assertion about observed exploitation activity, made by an observer at a given point in time. In an ideal world, the same actor that defines the vulnerability (for example, a vendor CNA) would also confirm exploitation, but in practice these concerns are frequently dissociated.

There are several common situations where vulnerability identity and exploitation assertions do not originate from the same place or even at the same time:

- Embargoed or restricted vulnerabilities, where exploitation is observed and tracked within a limited circle (CSIRTs, intelligence teams, trusted communities) before public disclosure or before a vendor-assigned identifier exists.
- Situations of disagreement or asymmetry of knowledge, where an observer has high-confidence evidence of exploitation while the vendor disputes impact, scope, or even the existence of the vulnerability.
- Early or partial observations, where exploitation activity is detected before a stable understanding of affected products, attack vectors, or vulnerability class has been established.

The KEV format explicitly models exploitation as an event-level assertion linked to a vulnerability identity, rather than as part of the identity itself. This reflects operational reality: exploitation is something that happens, may occur multiple times, may be observed by different parties, and may be interpreted differently as more information becomes available.

This separation already exists implicitly across today's KEV lists, vendor advisories, and threat intelligence reports, but it is inconsistently expressed, often duplicated, and rarely machine-parseable in a coherent way. The GCVE KEV format (BCP-07) provides a structured and open way to express these assertions while anchoring them to a shared vulnerability identifier when one exists.

Importantly, KEV entries are assertions, not ground truth.

They may later be revised, withdrawn, or contradicted by other observers. Decoupling identity from assertions allows such evolution without disconnecting from the underlying vulnerability record. A vulnerability identity should remain stable even if claims about exploitation change over time.

This model also assumes that multiple assertions per vulnerability identity are normal and expected. Different organisations may publish different KEV entries, CVSS scores, or detection signatures for the same GCVE ID, reflecting different perspectives, evidence sets, or confidence levels. Centralising identity while allowing plural assertions enables this diversity in a federated model.

With respect to identity-related data (such as vendor, product, or RCE status), once a vulnerability identity is established and assigned a GCVE ID, CVE ID (or any id), that identity should be the container for stable characteristics from a vendor. Assertions such as KEV entries, CVSS vectors, or signatures should primarily reference the GCVE ID, CVE ID (or any id) and focus on what is specific to the assertion itself: context, timing, confidence, evidence, or scoring rationale. The goal is not to repeat identity-defining attributes in every assertion.

Finally, this separation is essential for machine-readability. Clear boundaries between identity and assertions enable automated correlation, reasoning, and downstream decision-making across heterogeneous data sources. GCVE KEV (BCP-07) treats exploitation knowledge as structured data, without overloading the vulnerability identity layer itself.

3.1.1 Relationship between KEV and NKEV

BCP-07 distinguishes three separate concepts:

1. **Vulnerability identity** establishes what vulnerability is being referenced, for example by a GCVE, CVE, GHSA, or another vulnerability identifier.
2. **KEV assertion** states that a producer has observed or otherwise asserts exploitation of a vulnerability.

3. **NKEV assessment** records the result of evaluating a specific product or product version against explicitly identified vulnerability knowledge sources and determining whether any known vulnerability is exploitable in the assessed product context.

A KEV assertion and an NKEV assessment are therefore not logical opposites. The absence of a KEV assertion **MUST NOT** be interpreted as evidence that a product has no known exploitable vulnerabilities. Likewise, `status.exploited` set to `false` **MUST NOT** be interpreted as an NKEV statement. In a KEV assertion, `exploited: false` only describes the current state of that producer's exploitation assertion, for example following a withdrawal.

An NKEV assessment is always scoped and time-bounded. A result of `pass` means that no known exploitable vulnerability was identified **within the declared assessment scope, against the declared knowledge sources, and up to the declared knowledge cutoff time**. It does not mean that no vulnerability exists, that future exploitation is impossible, or that a product is automatically compliant with any legal or regulatory framework.

3.2 Format

It's a single JSON object (ECMA 404) per KEV entry. The KEV entry is associated to a vulnerability ID in GCVE ID or any known vulnerability identifier.

3.2.1 Sample

3.2.1.1 Combined KEV Assertion

The JSON file below provides an example of a KEV file referencing a GCVE vulnerability ID.

```
1 {
2   "vulnerability": {
3     "vulnId": "GCVE-0-2025-55182"
4   },
5   "status": {
6     "exploited": true,
7     "status_reason": "confirmed",
8     "status_updated_at": "2025-12-24T10:15:00Z"
9   },
10  "characteristics": {
11    "remote_code_execution": true,
12    "authentication_required": false,
13    "local_access_required": false
14  },
15  "timestamps": {
```

```
16     "first_seen_at": "2025-12-03T10:15:00Z",
17     "asserted_at": "2025-12-05T12:10:11Z",
18     "recorded_at": "2025-12-05T13:15:00Z",
19     "last_seen_at": "2025-12-24T09:42:21Z"
20 },
21 "scope": {
22     "observation_regions": ["Europe", "North America"],
23     "victim_countries": ["LU", "BE", "US", "CA"],
24     "sector": ["Telecoms", "Aerospace"],
25     "asset_exposure": ["internet-facing"],
26     "notes": "Regions reflect observed evidence, not global exclusivity
27     ."
28 },
29 "evidence": [
30     {
31         "type": "incident_response",
32         "signal": "confirmed_compromise",
33         "confidence": 0.9,
34         "source": "national-csirt",
35         "details": {
36             "observed_outcome": ["initial-access", "rce"],
37             "detection_basis": ["forensics", "log-analysis"]
38         }
39     },
40     {
41         "type": "honeypot",
42         "signal": "in_the_wild_attempts",
43         "confidence": 0.6,
44         "source": "research-honeynet",
45         "details": {
46             "attempt_volume": "high",
47             "successful_exploitation": false
48         }
49     }
50 ],
51 "references": [
52     {
53         "id": "GCVE-0-2025-55182",
54         "url": "https://vulnerability.circl.lu/vuln/CVE-2025-55182#
55         sightings"
56     }
57 ]
58 }
```

3.2.1.2 CISA KEV in BCP-07 Format

The JSON file below provides an example of a KEV file referencing a CISA KEV assertion.

```
1 {
```

```
2  "vulnerability": {
3    "vulnId": "CVE-2020-29583"
4  },
5  "status": {
6    "exploited": true,
7    "status_reason": "confirmed",
8    "status_updated_at": "2021-11-03T00:00:00Z"
9  },
10 "timestamps": {
11   "first_seen_at": "2021-11-03T00:00:00Z",
12   "asserted_at": "2021-11-03T00:00:00Z",
13   "recorded_at": "2026-01-22T05:07:44Z"
14 },
15 "evidence": [
16   {
17     "type": "vendor_report",
18     "signal": "successful_exploitation",
19     "confidence": 0.8,
20     "source": "cisa-kev",
21     "details": {
22       "feed": "CISA Known Exploited Vulnerabilities Catalog",
23       "date_added": "2021-11-03",
24       "due_date": "2022-05-03",
25       "vendorProject": "Zyxel",
26       "product": "Multiple Products",
27       "vulnerabilityName": "Zyxel Multiple Products Use of Hard-Coded
28         Credentials Vulnerability",
29       "knownRansomwareCampaignUse": "Unknown",
30       "cwes": [
31         "CWE-522"
32       ]
33     }
34   ],
35   "references": [
36     {
37       "id": "CVE-2020-29583",
38       "url": "https://www.cisa.gov/known-exploited-vulnerabilities-
39         catalog?search_api_fulltext=CVE-2020-29583"
40     }
41   ],
42   "scope": {
43     "notes": "KEV entry: Zyxel Multiple Products Use of Hard-Coded
44       Credentials Vulnerability | Affected: Zyxel / Multiple Products
45       | Description: Zyxel firewalls (ATP, USG, VM) and AP Controllers
46       (NXC2500 and NXC5500) contain a use of hard-coded credentials
47       vulnerability in an undocumented account (\"zyfwp\") with an
48       unchangeable password. | Required action: Apply updates per
49       vendor instructions. | Due date: 2022-05-03 | Known ransomware
50       campaign use (KEV): Unknown | Notes (KEV): https://nvd.nist.gov/
51       vuln/detail/CVE-2020-29583"
```

```
43   }
44 }
```

3.2.1.3 Withdrawn and Reasserted KEV Assertions

A producer can stop standing behind its own KEV assertion by setting `status.exploited` to **false**, setting `status.status_reason` to `withdrawn`, and setting `status.status_updated_at` to the time of withdrawal. All three fields **MUST** be present in a withdrawn assertion. A withdrawn assertion is a tombstone: it records that the producer previously made the assertion and subsequently withdrew it. It does not prove that exploitation never occurred. It is distinct from `disputed`, which represents an assessment that the exploitation claim is contested, and from `historical`, which means the producer assesses that exploitation did occur in the past.

```
1  {
2    "uuid": "1f47e98d-f850-4c2a-9e0f-c8dce5b0bdb1",
3    "vulnerability": {
4      "vulnId": "CVE-2026-69836"
5    },
6    "status": {
7      "exploited": false,
8      "status_reason": "withdrawn",
9      "status_updated_at": "2026-09-09T14:00:00Z"
10   },
11   "scope": {
12     "notes": "The asserting origin removed this vulnerability from its
13              KEV catalogue."
14   }
15 }
```

Producers **SHOULD** retain and publish a withdrawn assertion rather than delete it so that downstream mirrors can discover the withdrawal. When an assertion has a `uuid`, a producer **MUST** preserve that UUID during withdrawal. Evidence **MAY** be retained on a withdrawn record; when retained, it describes the evidence for the original claim and does not override the current `exploited: false` status.

Consumers **MUST** exclude every record with `status.exploited` set to **false** from the current KEV set by default, regardless of `status.status_reason`. The `withdrawn` reason additionally tells consumers to retain the record as a tombstone for audit and reconciliation. Interpretation and retention of other false-status records are consumer-defined.

`status.exploited: false` **MUST NOT** be used to express that a product has no known exploitable vulnerabilities. Product-level NKEV statements are represented by the separate NKEV assessment object defined later in this BCP.

If the same producer later stands behind the assertion again, including for a recurrent source catalogue, it can reassert the record by setting `status.exploited` to `true`, choosing the applicable non-`withdrawn status_reason`, and advancing `status.status_updated_at`. It **MUST** preserve the UUID when one exists. On reassertion, the producer **SHOULD** advance `timestamps.asserted_at` to the new assertion time and **SHOULD NOT** change `timestamps.first_seen_at` unless new evidence establishes an earlier first observation. The record can move between asserted and withdrawn states repeatedly. BCP-07 represents the current state of that assertion; transports or repositories that require a complete transition history should retain successive versions of the record.

3.2.2 Field Description

3.2.2.1 uuid

- **Type:** string (UUID version 4)
- **Required:** no
- **Description:** A globally unique identifier (UUID), version 4, assigned to the KEV record.

3.2.2.2 vulnerability Object

Describes the vulnerability being asserted as exploited.

3.2.2.2.1 vulnerability.vulnId

- **Type:** string
- **Required:** yes
- **Description:** GCVE, CVE identifier, GHSA or any identifier of the vulnerability.
- **Example:** "GCVE-0-2025-55182"

3.2.2.2.2 vulnerability.altId

- **Type:** array
- **Required:** no
- **Description:** Alternative identifiers that refer to the same vulnerability, used in addition to `vulnerability.vulnId`.

3.2.2.3 status Object

Represents the current exploitation status.

3.2.2.3.1 status.exploited

- **Type:** boolean
- **Description:** Indicates whether exploitation has been observed or asserted.
- **Semantics:** Does not imply global prevalence or universal exploitability.

3.2.2.3.2 status.status_reason

- **Type:** string (enum)
- **Allowed values:** `confirmed`, `suspected`, `disputed`, `historical`, `unknown`, `withdrawn`
- **Description:** Rationale behind the exploitation status.
- **Withdrawal semantics:** `withdrawn` means that the asserting origin no longer stands behind its own claim. When this value is used, `status.exploited` **MUST** be `false` and `status.status_updated_at` **MUST** be present. A later reassertion **MUST** set `status.exploited` to `true` and use an applicable non-`withdrawn` reason.

3.2.2.3.3 status.status_updated_at

- **Type:** string (RFC3339 datetime)
- **Description:** Timestamp of the last change to the exploitation status in the KEV assertion.

3.2.2.4 characteristics Object

Describes high-level technical characteristics of the vulnerability that are relevant to exploitation assessment, without providing exploit details or turning the KEV assertion into full threat intelligence.

These fields describe properties of the vulnerability itself, not necessarily every observed exploitation instance.

3.2.2.4.1 characteristics.remote_code_execution

- **Type:** boolean
- **Description:** Indicates whether successful exploitation can result in remote code execution.
- **Notes:** Does not imply exploit reliability or ease of weaponization.

3.2.2.4.2 `characteristics.authentication_required`

- **Type:** boolean
- **Description:** Indicates whether authentication is required to exploit the vulnerability.
- **Notes:** Reflects the weakest known exploitation path.

3.2.2.4.3 `characteristics.local_access_required`

- **Type:** boolean
- **Description:** Indicates whether local system access is required prior to exploitation.
- **Notes:** Useful to distinguish remote exploitation from post-compromise privilege escalation.

3.2.2.4.4 `characteristics.severity`

- **Type:** number (0.0–100)
- **Description:** Severity associated with this vulnerability. The `severity` field represents the severity inferred from observed exploitation activity, as asserted in the KEV record, rather than a vendor- or CVSS-derived vulnerability score. It reflects the operational impact demonstrated by exploitation evidence (e.g., full system compromise observed via telemetry or honeypot data).

3.2.2.5 `timestamps Object`

Separates different notions of time to avoid ambiguity.

3.2.2.5.1 `timestamps.first_seen_at`

- **Type:** string (RFC3339 datetime)
- **Description:** Earliest known exploitation activity based on technical observation.
- **Notes:** May be estimated and updated retroactively.

3.2.2.5.2 `timestamps.asserted_at`

- **Type:** string (RFC3339 datetime)
- **Description:** Date when an authority or source officially declared exploitation.
- **Notes:** Mirrors fields such as “date added” in KEV lists.

3.2.2.5.3 `timestamps.recorded_at`

- **Type:** string (RFC3339 datetime)

- **Description:** Timestamp when this assertion was ingested or recorded by the collector.
- **Notes:** System-specific and independent of the source.

3.2.2.5.4 `timestamps.last_seen_at`

- **Type:** string (RFC3339 datetime)
- **Description:** Most recent confirmed observation of exploitation activity.
- **Notes:** Optional and often unavailable.

3.2.2.6 `scope Object`

Defines the observed context of exploitation.

3.2.2.6.1 `scope.observation_regions`

- **Type:** array of strings
- **Description:** Geographic regions where exploitation evidence was observed. The region can be described in **UN M49** format to facilitate automation.
- **Notes:** Reflects sensor or reporting coverage, not global limits.

3.2.2.6.2 `scope.victim_countries`

- **Type:** array of strings
- **Description:** Countries in ISO 3166 where confirmed victims were identified.
- **Notes:** Often incomplete or unavailable.

3.2.2.6.3 `scope.sector`

- **Type:** array of strings
- **Description:** Sectors targeted or affected by exploitation. The sector **SHALL** come from the **MISP galaxy sector**.
- **Example:** "Telecoms", "Aerospace"

3.2.2.6.4 `scope.asset_exposure`

- **Type:** array of strings
- **Allowed values:** internet-facing, internal, vpn-accessible, unknown
- **Description:** Exposure context of affected assets.

3.2.2.6.5 scope.notes

- **Type:** string
- **Description:** Human-readable clarifications to prevent misinterpretation.

3.2.2.7 evidence Array

Collection of independent signals supporting the exploitation claim.

3.2.2.7.1 evidence[].type

- **Type:** string (enum)
- **Allowed values:** `incident_response`, `telemetry`, `honeypot`, `sinkhole`, `vendor_report`, `public_report`, `research_report`, `unknown`
- **Description:** Origin of the exploitation evidence.

3.2.2.7.2 evidence[].signal

- **Type:** string (enum)
- **Allowed values: (can be multiple)** `in_the_wild_attempts`, `successful_exploitation`, `confirmed_compromise`, `mass_scanning`, `weaponized_exploit_available`
- **Description:** Nature of the observed exploitation signal.

3.2.2.7.3 evidence[].confidence

- **Type:** number (0.0–1.0) or enum
- **Description:** Confidence level associated with this evidence.

The confidence **SHALL** come from the following confidence evaluation table to ensure consistent confidence values.

Confidence Label	Meaning (confidence in this evidence item)	Typical exploitation evidence examples
0.0 None	No usable evidence or placeholder only	Empty claim; unresolved rumor with no traceability
0.1 Extremely low	Unreliable and uncorroborated; major gaps	Single anonymous post; vague claim of exploitation

Confidence Label	Meaning (confidence in this evidence item)	Typical exploitation evidence examples
0.2 Very low	Weak signal; high chance of misattribution	Ambiguous scanning activity; noisy or indirect telemetry
0.3 Low	Plausible but thin; limited traceability	Single non-authoritative report; partial indicators without artifacts
0.4 Low–moderate	Some structure and traceability; still uncertain	Reputable researcher hint with limited technical detail
0.5 Moderate	Credible but not fully validated; alternative explanations remain	Multiple consistent reports; exploitation attempts seen but success unclear
0.6 Moderate–high	Good evidence with reasonable verification	Honeypot telemetry showing exploit-like behavior; strong IOCs
0.7 High	Strong and fairly direct evidence; limited uncertainty	Incident response report with logs/artifacts consistent with exploitation
0.8 Very high	Direct evidence or strong multi-source corroboration	Forensics confirming exploit path; authoritative confirmation of in-the-wild exploitation
0.9 Near-certain	Highly direct, well-attributed, well-corroborated	Confirmed compromise with clear exploit attribution across independent sources
1.0 Certain	Practically proven; no plausible alternative explanation	Deterministic proof (e.g., full packet/log capture) tying exploitation to the vulnerability

3.2.2.7.4 evidence[].source

- **Type:** string
- **Description:** Logical identifier of the reporting entity or data source. MISP org UUID? What about existing KEV source like CISA, ENISA or alike. Should we have an enum with existing ones?

The source would be the only required fields has many KEV like the type of signal.

3.2.2.7.5 `evidence[]`.`details`

- **Type:** object
- **Description:** Structured, free-form metadata describing how the signal was derived. Additional feeds from KEV sources which are not described in this format such as [cwes](#).
- **Notes:** Content is implementation-specific.

3.2.2.7.6 `evidence[]`.`gcve`

- **Type:** object
- **Description:** Structured object describing evidence originating from the GCVE ecosystem.
- **Required:** no

`gcve` Object

- `evidence[]`.`gcve`.`origin_uuid`
 - **Type:** string
 - **Description:** UUID of the origin instance where the assertion originated. GCVE maintains a list of **known KEVs** to determine the correct source UUID but the UUID **MUST** be consistent from the origin instance.
- `evidence[]`.`gcve`.`gna`
 - **Type:** number (0–65535)
 - **Description:** GNA ID identifying the origin of the assertion.
- `evidence[]`.`gcve`.`object_uuid`
 - **Type:** string
 - **Description:** UUID of the assertion associated with this evidence in the GCVE ecosystem.

3.2.2.8 `gcve` Object

- **Type:** object
- **Description:** Structured object describing evidence originating from the GCVE ecosystem at root level.
- **Required:** no

3.2.2.8.1 `gcve.origin_uuid`

- **Type:** string
- **Description:** UUID of the origin instance where the assertion originated. GCVE maintains a list of **known KEVs** to determine the correct source UUID but the UUID **MUST** be consistent from the origin instance.

3.2.2.8.2 `gcve.gna`

- **Type:** number (0–65535)
- **Description:** GNA ID identifying the origin of the assertion.

3.2.2.8.3 `gcve.object_uuid`

- **Type:** string
- **Description:** UUID of the assertion associated with this evidence in the GCVE ecosystem.

4 No Known Exploitable Vulnerability (NKEV) Assessment Format

The NKEV format describes a **product-level technical assessment**. It complements, but does not replace, KEV assertions.

A KEV assertion normally concerns one vulnerability and expresses that exploitation has been observed or asserted. An NKEV assessment starts from the opposite operational direction: it identifies a product or product version, defines the assessed configuration and scope, lists the vulnerability knowledge sources consulted, records relevant vulnerability findings and evidence, and produces a time-bounded technical result.

An NKEV assessment **MUST NOT** infer a `pass` result merely from the absence of BCP-07 KEV assertions. The assessment needs to consider the knowledge sources declared in `assessment.knowledge_sources` and determine applicability and exploitability for vulnerabilities relevant to the assessed product.

4.1 NKEV Assessment Object

The NKEV assessment is a separate JSON object from a KEV assertion. Implementations **SHOULD** use a stable `uuid` for an assessment when the record is expected to be updated, redistributed, or reconciled across systems.

The core object is stored below the root `assessment` key and has `type` set to `nkev`.

4.1.1 NKEV Result Semantics

`assessment.result` uses the following values:

- `pass`: no known exploitable vulnerability was identified within the declared scope, knowledge sources, configuration, and knowledge cutoff time.
- `fail`: at least one vulnerability was identified as applicable and exploitable in the assessed product context.

- **inconclusive**: the assessor could not reach a deterministic result, for example because product composition, source coverage, configuration, or exploitability evidence was incomplete.

These values represent **technical assessment results**. They do not by themselves express legal conformity, certification, or regulatory compliance.

4.1.2 Sample NKEV Assessment

```
1 {
2   "uuid": "3be2d7b8-8c6c-4a95-a8ad-4b95db4e8508",
3   "assessment": {
4     "type": "nkev",
5     "result": "pass",
6     "assessed_at": "2026-10-01T10:00:00Z",
7     "knowledge_cutoff_at": "2026-10-01T09:30:00Z",
8     "product": {
9       "vendor": "Example Corp",
10      "name": "Example Router",
11      "version": "4.2.1",
12      "identifiers": [
13        {
14          "type": "cpe",
15          "id": "cpe:2.3:h:example:router:4.2.1:*:*:*:*:*:*"
16        }
17      ]
18    },
19    "configurations": [
20      {
21        "state": "factory_default"
22      }
23    ],
24    "scope": {
25      "software": true,
26      "firmware": true,
27      "first_party": true,
28      "third_party": true,
29      "interfaces": true,
30      "remote_data_processing": false,
31      "notes": "Assessment covers the product version made available
32              with its factory-default configuration."
33    },
34    "knowledge_sources": [
35      {
36        "type": "vulnerability_database",
37        "name": "EUVD",
38        "url": "https://euvd.enisa.europa.eu/",
39        "queried_at": "2026-10-01T09:20:00Z"
```

```
39     },
40     {
41       "type": "vulnerability_database",
42       "name": "GCVE",
43       "url": "https://db.gcve.eu/",
44       "queried_at": "2026-10-01T09:21:00Z"
45     },
46     {
47       "type": "vendor_advisory",
48       "name": "Example Corp Security Advisories",
49       "url": "https://example.invalid/security/",
50       "queried_at": "2026-10-01T09:22:00Z"
51     }
52   ],
53   "findings": [
54     {
55       "vulnerability": {
56         "vulnId": "CVE-2026-12345"
57       },
58       "component": {
59         "name": "example-component",
60         "version": "1.2.3",
61         "identifiers": [
62           {
63             "type": "purl",
64             "id": "pkg:generic/example-component@1.2.3"
65           }
66         ]
67       },
68       "applicability": "affected",
69       "exploitability": "not_exploitable",
70       "justification": {
71         "type": "unreachable",
72         "description": "The affected function cannot be reached through any enabled product interface."
73       },
74       "evidence": [
75         {
76           "type": "vex_statement",
77           "source": "vendor",
78           "url": "https://example.invalid/security/vex/CVE-2026-12345.json"
79         }
80       ]
81     }
82   ],
83   "evidence": [
84     {
85       "type": "software_composition_analysis",
86       "source": "assessment-toolchain",
87       "tool": {
```

```
88         "name": "ExampleSCA",
89         "version": "2.4"
90     },
91 },
92 {
93     "type": "authenticated_scan",
94     "source": "assessment-toolchain",
95     "tool": {
96         "name": "ExampleScanner",
97         "version": "5.1"
98     }
99 }
100 ]
101 },
102 "gcve": {
103     "origin_uuid": "1a89b78e-f703-45f3-bb86-59eb712668bd",
104     "gna": 1,
105     "object_uuid": "3be2d7b8-8c6c-4a95-a8ad-4b95db4e8508"
106 }
107 }
```

4.2 NKEV Field Description

4.2.1 uuid

- **Type:** string (UUID version 4)
- **Required:** no
- **Description:** Stable identifier assigned to the NKEV assessment record.

4.2.2 assessment Object

- **Type:** object
- **Required:** yes
- **Description:** Contains the NKEV assessment itself.

4.2.2.1 assessment.type

- **Type:** string
- **Required:** yes
- **Allowed value:** `nkev`
- **Description:** Identifies the object as an NKEV assessment.

4.2.2.2 `assessment.result`

- **Type:** string (enum)
- **Required:** yes
- **Allowed values:** `pass`, `fail`, `inconclusive`
- **Description:** Overall technical result of the NKEV assessment.

A `pass` result is valid only for the declared product, configuration, scope, knowledge sources, and assessment time. It **MUST NOT** be interpreted as a timeless statement or as a legal conformity decision.

4.2.2.3 `assessment.assessed_at`

- **Type:** string (RFC3339 datetime)
- **Required:** yes
- **Description:** Time at which the assessment result was produced.

4.2.2.4 `assessment.knowledge_cutoff_at`

- **Type:** string (RFC3339 datetime)
- **Required:** yes
- **Description:** Latest point in time up to which vulnerability knowledge was considered by the assessment.

The knowledge cutoff makes the time-bounded nature of NKEV explicit. Consumers **MUST NOT** assume that a `pass` result remains valid after this timestamp without considering newer vulnerability knowledge.

4.2.3 `assessment.product` Object

Identifies the product or product version being assessed.

4.2.3.1 `assessment.product.vendor`

- **Type:** string
- **Required:** no
- **Description:** Product vendor or project name.

4.2.3.2 `assessment.product.name`

- **Type:** string
- **Required:** yes
- **Description:** Product name.

4.2.3.3 `assessment.product.version`

- **Type:** string
- **Required:** yes
- **Description:** Product version being assessed.

4.2.3.4 `assessment.product.identifiers`

- **Type:** array
- **Required:** no
- **Description:** Machine-readable identifiers for the product or product version.

Each entry contains:

- `type`: identifier type such as `cpe`, `purl`, `swid`, `vendor`, or `other`.
- `id`: identifier value.

CPE or PURL are useful where available but neither is mandatory.

4.2.4 `assessment.configurations` Array

- **Type:** array
- **Required:** yes
- **Description:** Configuration states covered by the assessment.

`assessment.configurations[]`.`state` uses the following values:

- `factory_default`
- `documented_configuration`
- `deployment_specific`
- `factory_reset`
- `unknown`

An optional `identifier` can name a documented configuration or deployment profile, and an optional `description` can provide additional context.

4.2.5 `assessment.scope` Object

Defines the product boundary covered by the assessment.

The following boolean fields can be used:

- `software`: software components are covered.
- `firmware`: firmware components are covered.
- `first_party`: first-party components are covered.
- `third_party`: third-party components are covered.
- `interfaces`: product interfaces are covered.
- `remote_data_processing`: remote data processing or remote service dependencies are covered where applicable.

`notes` can describe additional scope limitations or assumptions.

An NKEV assessment **SHOULD** cover the whole assessed product and should not limit itself to third-party components listed in an SBOM when first-party software, firmware, interfaces, or other product elements can contain vulnerabilities.

4.2.6 `assessment.knowledge_sources` Array

- **Type:** array
- **Required:** yes
- **Description:** Vulnerability knowledge sources consulted during the assessment.

Each source contains:

- `type`: one of `vulnerability_database`, `kev_catalogue`, `vendor_advisory`, `upstream_advisory`, `csirt_advisory`, `research_source`, or `other`.
- `name`: human-readable source name.
- `url`: optional URI identifying the source.
- `queried_at`: RFC3339 timestamp recording when the source was consulted.
- `scope`: optional description of which components or product areas the source was used for.

An NKEV result is only meaningful in relation to its declared knowledge sources. Implementations **SHOULD** use sources appropriate to the product category and **MAY** include BCP-07 KEV catalogues as one input source.

4.2.7 `assessment.findings` Array

Contains vulnerability-specific findings discovered during the assessment. A `pass` assessment can still contain findings when vulnerabilities were matched to the product but were determined to be not applicable or not exploitable in the assessed context.

4.2.7.1 `assessment.findings[].vulnerability`

- **Type:** object
- **Required:** yes
- **Description:** Vulnerability identity associated with the finding.

It uses the same `vulnId` and optional `altId` model as the KEV assertion format.

4.2.7.2 `assessment.findings[].component`

- **Type:** object
- **Required:** no
- **Description:** Product component associated with the finding.

The component can contain `name`, `version`, and an `identifiers` array using the same identifier object as `assessment.product.identifiers`.

4.2.7.3 `assessment.findings[].applicability`

- **Type:** string (enum)
- **Required:** yes
- **Allowed values:** `affected`, `not_affected`, `unknown`
- **Description:** Whether the vulnerability applies to the product or component version under assessment.

4.2.7.4 `assessment.findings[].exploitability`

- **Type:** string (enum)
- **Required:** yes
- **Allowed values:** `exploitable`, `not_exploitable`, `under_investigation`, `unknown`
- **Description:** Product-specific exploitability determination.

Applicability and exploitability are distinct. A component can match an affected version while exploitation is prevented by a verifiable property of the assessed product. Conversely, a finding with `applicability: not_affected` is outside the affected product/version range even before exploitability is considered.

4.2.7.5 `assessment.findings[].justification`

- **Type:** object
- **Required:** no
- **Description:** Structured justification for an applicability or exploitability determination.

`justification.type` can use:

- `affected_code_absent`
- `affected_function_absent`
- `component_not_present`
- `version_not_affected`
- `unreachable`
- `mitigated_by_default`
- `operational_conditions_not_met`
- `false_positive`
- `other`

A human-readable `description` **SHOULD** be provided when the reason is not self-explanatory or when `type` is `other`.

For `not_exploitable` findings, the justification **SHOULD** be based on an objectively verifiable product property or condition. Merely tracking a vulnerability for future remediation is not, by itself, evidence that the vulnerability is not exploitable.

4.2.8 `assessment.findings[].evidence` and `assessment.evidence`

NKEV evidence is intentionally separate from the exploitation evidence used by KEV assertions.

`assessment.findings[].evidence` describes evidence supporting one vulnerability-specific finding. `assessment.evidence` describes evidence supporting the assessment as a whole.

Allowed evidence types include:

- `sbom`
- `binary_analysis`

- `software_composition_analysis`
- `vulnerability_scan`
- `authenticated_scan`
- `unauthenticated_scan`
- `source_code_analysis`
- `configuration_analysis`
- `vendor_statement`
- `vex_statement`
- `manual_analysis`
- `test_result`
- `bcp07_kev_assertion`
- `other`

An evidence item can contain:

- `source`: logical identifier of the producer or assessment source.
- `url`: optional URI for external evidence.
- `tool`: optional object with `name` and `version`.
- `reference`: optional object for referencing another record, for example a BCP-07 KEV assertion by `uuid` or URI.
- `details`: implementation-specific structured metadata.

An SBOM or component inventory **MAY** be used as assessment evidence, but the existence of an SBOM **MUST NOT** by itself be interpreted as evidence that the product contains no known exploitable vulnerabilities.

4.2.9 Referencing BCP-07 KEV Assertions from NKEV

An NKEV assessment can use one or more BCP-07 KEV assertions as evidence that exploitation has been observed for a vulnerability. Such references should use an evidence item with `type` set to `bcp07_kev_assertion`.

```
1 {
2   "type": "bcp07_kev_assertion",
3   "source": "gcve-kev-catalogue",
4   "reference": {
5     "uuid": "1f47e98d-f850-4c2a-9e0f-c8dce5b0bdb1",
6     "url": "https://example.invalid/kev/1f47e98d-f850-4c2a-9e0f-c8dce5b0bdb1"
7   }
8 }
```

A referenced KEV assertion is one input to the assessment. It does not by itself establish that the vulnerability is applicable or exploitable in the specific product being assessed; applicability and product-specific exploitability still need to be determined.

4.2.10 Interoperability with VEX and Other Vulnerability Formats

BCP-07 does not require a specific vulnerability advisory or product-security format. NKEV findings can reference or be derived from VEX-style statements, including machine-readable statements that distinguish affected, not affected, and product-specific exploitability conditions. Such external statements are evidence inputs; the NKEV assessment remains an independent BCP-07 object with its own scope, time reference, knowledge sources, and result.

4.3 GCVE KEV Directory and Source Index Format

BCP-07 describes individual KEV assertions, but implementers also need a way to discover which organisations publish KEV catalogues and where their machine-readable feeds are located. The GCVE initiative therefore maintains the [GCVE KEV directory](#), a non-mandatory index in which KEV producers can announce their catalogues and exploitation assertions.

A KEV producer can be a vendor, a CSIRT, or any other organisation observing or reporting exploitation. Producers register a stable source entry in the directory and provide one of the following machine-readable endpoints:

- A `bcp_07_url` that directly publishes assertions in the BCP-07 format defined by this document.
- An `automation_url` that publishes a native KEV format. Where direct BCP-07 output is not available, a converter can transform that feed; the [gcve-eu-kev project](#) provides scripts and examples for such conversions.

Consumers and catalogue operators can use the directory to discover these distributed sources, retrieve or convert their assertions, and combine them into a more complete overview without treating any single producer as the universal source of truth. One implementation of this aggregation workflow is the [GCVE KEV catalogues overview](#), provided by Vulnerability-Lookup.

The index is a catalogue of KEV catalogues. It is not authoritative for the content of any listed catalogue and it is not required for a producer to publish BCP-07 KEV assertions. Its purpose is to facilitate discovery, correlation, and stable attribution of KEV catalogues, especially where `gcve.origin_uuid` or `evidence[] .gcve.origin_uuid` values need to be resolved to a known source.

4.3.1 Index Object

The index is a single JSON object (ECMA 404) with a `kev` array. Each item in the array describes one KEV catalogue or source.

```
1 {
2   "kev": [
3     {
4       "uuid": "1a89b78e-f703-45f3-bb86-59eb712668bd",
5       "short_name": "CIRCL",
6       "gcve_gna_id": 1,
7       "url": "https://vulnerability.circl.lu/known-exploited-
          vulnerabilities-catalog",
8       "automation_url": "https://vulnerability.circl.lu/api/kev/?
          vulnerability_lookup_origin=1a89b78e-f703-45f3-bb86-59
          eb712668bd",
9       "bcp_07_url": "https://vulnerability.circl.lu/api/kev/bcp-07/?
          vulnerability_lookup_origin=1a89b78e-f703-45f3-bb86-59
          eb712668bd",
10      "description": "CIRCL provides a known-exploited vulnerability
          catalog and supports the different status reasons described in
          GCVE BCP-07.",
11      "meta": {
12        "license": "CC-BY-4.0"
13      }
14    }
15  ]
16 }
```

4.3.2 Index Field Description

4.3.2.1 `kev` Array

- **Type:** array
- **Required:** yes
- **Description:** Collection of KEV catalogue source descriptions.

4.3.2.2 `kev[].uuid`

- **Type:** string (UUID)
- **Required:** yes
- **Description:** Stable UUID identifying the KEV catalogue source. When a BCP-07 assertion uses `gcve.origin_uuid` or `evidence[].gcve.origin_uuid`, the value should match this UUID where the source is listed in the index.

4.3.2.3 `kev[] .short_name`

- **Type:** string
- **Required:** yes
- **Description:** Short human-readable name for the KEV catalogue source.

4.3.2.4 `kev[] .gcve_gna_id`

- **Type:** number (0–65535)
- **Required:** no
- **Description:** GNA ID associated with the source, when the source is a GCVE Numbering Authority or otherwise has an assigned GNA identifier.

4.3.2.5 `kev[] .url`

- **Type:** string (URI)
- **Required:** yes
- **Description:** Human-facing URL for the KEV catalogue or source.

4.3.2.6 `kev[] .automation_url`

- **Type:** string (URI)
- **Required:** no
- **Description:** Machine-readable URL for the source catalogue in its native or preferred automation format. This field may point to a format other than BCP-07.

4.3.2.7 `kev[] .bcp_07_url`

- **Type:** string (URI)
- **Required:** no
- **Description:** Machine-readable URL where the KEV data can be retrieved in BCP-07 format. This field is optional because not every KEV source publishes BCP-07 directly. When present, consumers can prefer this URL over `automation_url` when they specifically need BCP-07 assertions.

4.3.2.8 `kev[].description`

- **Type:** string
- **Required:** no
- **Description:** Human-readable description of the KEV catalogue source.

4.3.2.9 `kev[].meta`

- **Type:** object
- **Required:** no
- **Description:** Optional metadata about the KEV catalogue source. This object is intentionally extensible to allow source-specific metadata without changing the core index format.

4.3.2.9.1 `kev[].meta.license`

- **Type:** string
- **Required:** no
- **Description:** Licence identifier for the KEV catalogue when available. SPDX identifiers SHOULD be used where an SPDX identifier exists, for example `CC-BY-4.0`, `CC0-1.0`, or `Apache-2.0`.

4.3.3 JSON Schema for the GCVE KEV Directory Array

The following schema validates only the array of source objects stored below the directory's top-level `kev` key. It does not validate the enclosing directory object.

```
1 {
2   "$schema": "https://json-schema.org/draft/2020-12/schema",
3   "$id": "https://gcve.eu/schemas/bcp-07-kev-directory-array.schema.json",
4   "title": "GCVE-BCP-07 KEV Directory Array",
5   "description": "The array of KEV catalogue sources stored below the directory's kev key.",
6   "type": "array",
7   "items": {
8     "type": "object",
9     "additionalProperties": false,
10    "required": [
11      "uuid",
12      "short_name",
13      "url"
14    ],
15    "properties": {
16      "uuid": {
```

```
17     "type": "string",
18     "format": "uuid",
19     "description": "Stable UUID identifying the KEV catalogue
    source."
20 },
21 "short_name": {
22     "type": "string",
23     "description": "Short human-readable name for the KEV catalogue
    source."
24 },
25 "gcve_gna_id": {
26     "type": "integer",
27     "minimum": 0,
28     "maximum": 65535,
29     "description": "GNA ID associated with the source, when
    available."
30 },
31 "url": {
32     "type": "string",
33     "format": "uri",
34     "description": "Human-facing URL for the KEV catalogue or
    source."
35 },
36 "automation_url": {
37     "type": "string",
38     "format": "uri",
39     "description": "Machine-readable URL for the source catalogue
    in its native or preferred automation format."
40 },
41 "bcp_07_url": {
42     "type": "string",
43     "format": "uri",
44     "description": "Machine-readable URL where KEV data can be
    retrieved in BCP-07 format."
45 },
46 "description": {
47     "type": "string",
48     "description": "Human-readable description of the KEV catalogue
    source."
49 },
50 "meta": {
51     "type": "object",
52     "description": "Optional extensible metadata about the KEV
    catalogue source.",
53     "additionalProperties": true,
54     "properties": {
55         "license": {
56             "type": "string",
57             "description": "Licence identifier for the KEV catalogue;
    use an SPDX identifier when available."
58         }
59     }
60 }
```

```
59     }  
60   }  
61 }  
62 }  
63 }
```

4.3.4 JSON Schema - KEV Assertion

JSON Schema - GCVE-BCP-07 Known Exploited Vulnerability (KEV) Assertion Format.

```
1  {  
2    "$schema": "https://json-schema.org/draft/2020-12/schema",  
3    "$id": "https://gcve.eu/schemas/bcp-07-kev-assertion.schema.json",  
4    "title": "GCVE-BCP-07 Known Exploited Vulnerability (KEV) Assertion",  
5    "type": "object",  
6    "additionalProperties": false,  
7    "required": [  
8      "vulnerability",  
9      "status"  
10   ],  
11   "properties": {  
12     "vulnerability": {  
13       "type": "object",  
14       "additionalProperties": false,  
15       "required": [  
16         "vulnId"  
17       ],  
18       "properties": {  
19         "vulnId": {  
20           "type": "string",  
21           "description": "GCVE, CVE, GHSA or any identifier of the  
22             vulnerability."  
23         },  
24         "altId": {  
25           "type": "array",  
26           "description": "Alternative identifiers that refer to the  
27             same vulnerability, used in addition to vulnerability.  
28             vulnId.",  
29           "items": {  
30             "type": "string"  
31           }  
32         }  
33       }  
34     },  
35     "gcve": {  
36       "$ref": "#/$defs/gcveRoot",  
37       "description": "Structured object describing GCVE origin metadata  
38         for the KEV assertion."  
39     }  
40   }  
41 }
```

```
36     "uuid": {
37         "type": "string",
38         "format": "uuid",
39         "description": "Globally unique identifier (UUID v4) assigned to
         the KEV assertion record."
40     },
41     "status": {
42         "type": "object",
43         "additionalProperties": false,
44         "allOf": [
45             {
46                 "if": {
47                     "properties": {
48                         "status_reason": {
49                             "const": "withdrawn"
50                         }
51                     },
52                     "required": [
53                         "status_reason"
54                     ]
55                 },
56                 "then": {
57                     "properties": {
58                         "exploited": {
59                             "const": false
60                         }
61                     },
62                     "required": [
63                         "exploited",
64                         "status_updated_at"
65                     ]
66                 }
67             }
68         ],
69         "properties": {
70             "exploited": {
71                 "type": "boolean",
72                 "description": "Indicates whether exploitation has been
                     observed or asserted."
73             },
74             "status_reason": {
75                 "type": "string",
76                 "description": "Rationale behind the exploitation status.",
77                 "enum": [
78                     "confirmed",
79                     "suspected",
80                     "disputed",
81                     "historical",
82                     "unknown",
83                     "withdrawn"
84                 ]
85             }
86         }
87     }
88 }
```

```
85     },
86     "status_updated_at": {
87         "type": "string",
88         "format": "date-time",
89         "description": "Timestamp of the last change to the
90             exploitation status in the KEV assertion (RFC3339).\"
91     }
92 },
93 "characteristics": {
94     "type": "object",
95     "additionalProperties": false,
96     "description": "High-level technical characteristics relevant to
97         exploitation assessment.\",
98     "properties": {
99         "remote_code_execution": {
100             "type": "boolean",
101             "description": "Whether successful exploitation can result in
102                 remote code execution.\"
103         },
104         "authentication_required": {
105             "type": "boolean",
106             "description": "Whether authentication is required to exploit
107                 the vulnerability.\"
108         },
109         "local_access_required": {
110             "type": "boolean",
111             "description": "Whether local system access is required prior
112                 to exploitation.\"
113         },
114         "severity": {
115             "type": "number",
116             "minimum": 0,
117             "maximum": 100,
118             "description": "Severity associated with this vulnerability -
119                 (0.0100).\"
120         }
121     }
122 },
123 "timestamps": {
124     "type": "object",
125     "additionalProperties": false,
126     "description": "Separate notions of time to avoid ambiguity.\",
127     "properties": {
128         "first_seen_at": {
129             "type": "string",
130             "format": "date-time",
131             "description": "Earliest known exploitation activity based on
132                 technical observation (RFC3339).\"
133         },
134         "asserted_at": {
```

```
129         "type": "string",
130         "format": "date-time",
131         "description": "Date when an authority or source officially
                        declared exploitation (RFC3339).\"
132     },
133     \"recorded_at\": {
134         \"type\": \"string\",
135         \"format\": \"date-time\",
136         \"description\": \"Timestamp when this assertion was ingested/
                        recorded by the collector (RFC3339).\"
137     },
138     \"last_seen_at\": {
139         \"type\": \"string\",
140         \"format\": \"date-time\",
141         \"description\": \"Most recent confirmed observation of
                        exploitation activity (RFC3339).\"
142     }
143 }
144 },
145 \"scope\": {
146     \"type\": \"object\",
147     \"additionalProperties\": false,
148     \"description\": \"Observed context of exploitation.\",
149     \"properties\": {
150         \"observation_regions\": {
151             \"type\": \"array\",
152             \"description\": \"Geographic regions where exploitation
                            evidence was observed (optionally UN M49).\",
153             \"items\": {
154                 \"type\": \"string\"
155             }
156         },
157         \"victim_countries\": {
158             \"type\": \"array\",
159             \"description\": \"Countries (ISO 3166) where confirmed victims
                            were identified.\",
160             \"items\": {
161                 \"type\": \"string\",
162                 \"minLength\": 2,
163                 \"maxLength\": 2
164             }
165         },
166         \"sector\": {
167             \"type\": \"array\",
168             \"description\": \"Sectors targeted/affected (SHALL come from
                            MISP galaxy sector).\",
169             \"items\": {
170                 \"type\": \"string\"
171             }
172         },
173         \"asset_exposure\": {
```

```
174         "type": "array",
175         "description": "Exposure context of affected assets.",
176         "items": {
177             "type": "string",
178             "enum": [
179                 "internet-facing",
180                 "internal",
181                 "vpn-accessible",
182                 "unknown"
183             ]
184         }
185     },
186     "notes": {
187         "type": "string",
188         "description": "Human-readable clarifications to prevent
189             misinterpretation."
189     }
190 },
191 },
192 "evidence": {
193     "type": "array",
194     "description": "Collection of independent signals supporting the
195         exploitation claim.",
196     "items": {
197         "$ref": "#/$defs/evidenceItem"
198     }
199 },
200 "references": {
201     "type": "array",
202     "description": "Links/IDs referencing external resources about
203         the vulnerability or sightings.",
204     "items": {
205         "$ref": "#/$defs/reference"
206     }
207 },
208 "$defs": {
209     "reference": {
210         "type": "object",
211         "additionalProperties": false,
212         "required": [
213             "id",
214             "url"
215         ],
216         "properties": {
217             "id": {
218                 "type": "string"
219             },
220             "url": {
221                 "type": "string",
222                 "format": "uri"
```

```
222     }
223   }
224 },
225 "confidence": {
226   "description": "Confidence level: number -(0.01.0) or an
227   implementation-specific enum/string.",
228   "oneOf": [
229     {
230       "type": "number",
231       "minimum": 0,
232       "maximum": 1
233     },
234     {
235       "type": "string"
236     }
237   ],
238   "evidenceSignal": {
239     "oneOf": [
240       {
241         "type": "string",
242         "enum": [
243           "in_the_wild_attempts",
244           "successful_exploitation",
245           "confirmed_compromise",
246           "mass_scanning",
247           "weaponized_exploit_available"
248         ]
249       },
250       {
251         "type": "array",
252         "items": {
253           "type": "string",
254           "enum": [
255             "in_the_wild_attempts",
256             "successful_exploitation",
257             "confirmed_compromise",
258             "mass_scanning",
259             "weaponized_exploit_available"
260           ]
261         },
262         "minItems": 1,
263         "uniqueItems": true
264       }
265     ]
266   },
267   "gcveEvidence": {
268     "type": "object",
269     "additionalProperties": false,
270     "properties": {
271       "origin_uuid": {
```

```
272         "type": "string",
273         "description": "UUID of the origin instance where the
           assertion originated."
274     },
275     "gna": {
276         "type": "integer",
277         "minimum": 0,
278         "maximum": 65535,
279         "description": "GNA ID identifying the origin of the
           assertion."
280     },
281     "object_uuid": {
282         "type": "string",
283         "description": "UUID of the assertion associated with this
           evidence in the GCVE ecosystem."
284     }
285 }
286 },
287 "gcveRoot": {
288     "type": "object",
289     "additionalProperties": false,
290     "description": "GCVE metadata describing the origin and identity
           of the KEV assertion.",
291     "properties": {
292         "origin_uuid": {
293             "type": "string",
294             "format": "uuid",
295             "description": "UUID of the origin instance where the KEV
           assertion originated."
296         },
297         "gna": {
298             "type": "integer",
299             "minimum": 0,
300             "maximum": 65535,
301             "description": "GNA ID identifying the origin of the KEV
           assertion."
302         },
303         "object_uuid": {
304             "type": "string",
305             "format": "uuid",
306             "description": "UUID of the KEV assertion in the GCVE
           ecosystem."
307         }
308     }
309 },
310 "evidenceItem": {
311     "type": "object",
312     "additionalProperties": false,
313     "required": [
314         "source"
315     ],

```

```
316     "properties": {
317         "type": {
318             "type": "string",
319             "description": "Origin of the exploitation evidence.",
320             "enum": [
321                 "incident_response",
322                 "telemetry",
323                 "honeypot",
324                 "sinkhole",
325                 "vendor_report",
326                 "csirt_report",
327                 "public_report",
328                 "research_report",
329                 "unknown"
330             ]
331         },
332         "signal": {
333             "$ref": "#/$defs/evidenceSignal",
334             "description": "Nature of the observed exploitation signal (
335                 string or array of strings)."
336         },
337         "confidence": {
338             "$ref": "#/$defs/confidence"
339         },
340         "source": {
341             "type": "string",
342             "description": "Logical identifier of the reporting entity or
343                 data source."
344         },
345         "details": {
346             "type": "object",
347             "description": "Structured, free-form metadata describing how
348                 the signal was derived (implementation-specific).",
349             "additionalProperties": true
350         },
351         "gcve": {
352             "$ref": "#/$defs/gcveEvidence",
353             "description": "Structured object describing evidence
354                 originating from the GCVE ecosystem."
355         }
356     }
357 }
```

4.3.5 JSON Schema - NKEV Assessment

JSON Schema - GCVE-BCP-07 No Known Exploitable Vulnerability (NKEV) Assessment Format.

```
1 {
2   "$schema": "https://json-schema.org/draft/2020-12/schema",
3   "$id": "https://gcve.eu/schemas/bcp-07-nkev-assessment.schema.json",
4   "title": "GCVE-BCP-07 No Known Exploitable Vulnerability (NKEV)
5     Assessment",
6   "type": "object",
7   "additionalProperties": false,
8   "required": [
9     "assessment"
10  ],
11  "properties": {
12    "uuid": {
13      "type": "string",
14      "format": "uuid",
15      "description": "Stable UUID assigned to the NKEV assessment
16        record."
17    },
18    "assessment": {
19      "$ref": "#/$defs/assessment"
20    },
21    "gcve": {
22      "$ref": "#/$defs/gcveRoot"
23    },
24    "references": {
25      "type": "array",
26      "items": {
27        "$ref": "#/$defs/reference"
28      }
29    }
30  },
31  "$defs": {
32    "assessment": {
33      "type": "object",
34      "additionalProperties": false,
35      "required": [
36        "type",
37        "result",
38        "assessed_at",
39        "knowledge_cutoff_at",
40        "product",
41        "configurations",
42        "knowledge_sources"
43      ],
44      "properties": {
45        "type": {
46          "const": "nkev"
47        },
48        "result": {
49          "type": "string",
50          "enum": [
51            "pass",
```

```
50         "fail",
51         "inconclusive"
52     ]
53 },
54 "assessed_at": {
55     "type": "string",
56     "format": "date-time"
57 },
58 "knowledge_cutoff_at": {
59     "type": "string",
60     "format": "date-time"
61 },
62 "product": {
63     "$ref": "#/$defs/product"
64 },
65 "configurations": {
66     "type": "array",
67     "minItems": 1,
68     "items": {
69         "$ref": "#/$defs/configuration"
70     }
71 },
72 "scope": {
73     "$ref": "#/$defs/assessmentScope"
74 },
75 "knowledge_sources": {
76     "type": "array",
77     "minItems": 1,
78     "items": {
79         "$ref": "#/$defs/knowledgeSource"
80     }
81 },
82 "findings": {
83     "type": "array",
84     "items": {
85         "$ref": "#/$defs/finding"
86     }
87 },
88 "evidence": {
89     "type": "array",
90     "items": {
91         "$ref": "#/$defs/assessmentEvidence"
92     }
93 }
94 }
95 },
96 "product": {
97     "type": "object",
98     "additionalProperties": false,
99     "required": [
100         "name",
```

```
101     "version"
102   ],
103   "properties": {
104     "vendor": {
105       "type": "string"
106     },
107     "name": {
108       "type": "string"
109     },
110     "version": {
111       "type": "string"
112     },
113     "identifiers": {
114       "type": "array",
115       "items": {
116         "$ref": "#/$defs/identifier"
117       }
118     }
119   }
120 },
121 "component": {
122   "type": "object",
123   "additionalProperties": false,
124   "properties": {
125     "name": {
126       "type": "string"
127     },
128     "version": {
129       "type": "string"
130     },
131     "identifiers": {
132       "type": "array",
133       "items": {
134         "$ref": "#/$defs/identifier"
135       }
136     }
137   }
138 },
139 "identifier": {
140   "type": "object",
141   "additionalProperties": false,
142   "required": [
143     "type",
144     "id"
145   ],
146   "properties": {
147     "type": {
148       "type": "string",
149       "enum": [
150         "cpe",
151         "purl",
```

```
152         "swid",
153         "vendor",
154         "other"
155     ]
156 },
157     "id": {
158         "type": "string"
159     }
160 },
161 },
162     "configuration": {
163         "type": "object",
164         "additionalProperties": false,
165         "required": [
166             "state"
167         ],
168         "properties": {
169             "state": {
170                 "type": "string",
171                 "enum": [
172                     "factory_default",
173                     "documented_configuration",
174                     "deployment_specific",
175                     "factory_reset",
176                     "unknown"
177                 ]
178             },
179             "identifier": {
180                 "type": "string"
181             },
182             "description": {
183                 "type": "string"
184             }
185         }
186 },
187     "assessmentScope": {
188         "type": "object",
189         "additionalProperties": false,
190         "properties": {
191             "software": {
192                 "type": "boolean"
193             },
194             "firmware": {
195                 "type": "boolean"
196             },
197             "first_party": {
198                 "type": "boolean"
199             },
200             "third_party": {
201                 "type": "boolean"
202             },
```

```
203     "interfaces": {
204         "type": "boolean"
205     },
206     "remote_data_processing": {
207         "type": "boolean"
208     },
209     "notes": {
210         "type": "string"
211     }
212 }
213 },
214 "knowledgeSource": {
215     "type": "object",
216     "additionalProperties": false,
217     "required": [
218         "type",
219         "name",
220         "queried_at"
221     ],
222     "properties": {
223         "type": {
224             "type": "string",
225             "enum": [
226                 "vulnerability_database",
227                 "kev_catalogue",
228                 "vendor_advisory",
229                 "upstream_advisory",
230                 "csirt_advisory",
231                 "research_source",
232                 "other"
233             ]
234         },
235         "name": {
236             "type": "string"
237         },
238         "url": {
239             "type": "string",
240             "format": "uri"
241         },
242         "queried_at": {
243             "type": "string",
244             "format": "date-time"
245         },
246         "scope": {
247             "type": "string"
248         }
249     }
250 },
251 "vulnerability": {
252     "type": "object",
253     "additionalProperties": false,
```

```
254     "required": [  
255         "vulnId"  
256     ],  
257     "properties": {  
258         "vulnId": {  
259             "type": "string"  
260         },  
261         "altId": {  
262             "type": "array",  
263             "items": {  
264                 "type": "string"  
265             }  
266         }  
267     }  
268 },  
269 "finding": {  
270     "type": "object",  
271     "additionalProperties": false,  
272     "required": [  
273         "vulnerability",  
274         "applicability",  
275         "exploitability"  
276     ],  
277     "properties": {  
278         "vulnerability": {  
279             "$ref": "#/$defs/vulnerability"  
280         },  
281         "component": {  
282             "$ref": "#/$defs/component"  
283         },  
284         "applicability": {  
285             "type": "string",  
286             "enum": [  
287                 "affected",  
288                 "not_affected",  
289                 "unknown"  
290             ]  
291         },  
292         "exploitability": {  
293             "type": "string",  
294             "enum": [  
295                 "exploitable",  
296                 "not_exploitable",  
297                 "under_investigation",  
298                 "unknown"  
299             ]  
300         },  
301         "justification": {  
302             "$ref": "#/$defs/justification"  
303         },  
304         "evidence": {
```

```
305         "type": "array",
306         "items": {
307             "$ref": "#/$defs/assessmentEvidence"
308         }
309     }
310 },
311 },
312 "justification": {
313     "type": "object",
314     "additionalProperties": false,
315     "required": [
316         "type"
317     ],
318     "properties": {
319         "type": {
320             "type": "string",
321             "enum": [
322                 "affected_code_absent",
323                 "affected_function_absent",
324                 "component_not_present",
325                 "version_not_affected",
326                 "unreachable",
327                 "mitigated_by_default",
328                 "operational_conditions_not_met",
329                 "false_positive",
330                 "other"
331             ]
332         },
333         "description": {
334             "type": "string"
335         }
336     }
337 },
338 "assessmentEvidence": {
339     "type": "object",
340     "additionalProperties": false,
341     "required": [
342         "type",
343         "source"
344     ],
345     "properties": {
346         "type": {
347             "type": "string",
348             "enum": [
349                 "sbom",
350                 "binary_analysis",
351                 "software_composition_analysis",
352                 "vulnerability_scan",
353                 "authenticated_scan",
354                 "unauthenticated_scan",
355                 "source_code_analysis",
```

```
356         "configuration_analysis",
357         "vendor_statement",
358         "vex_statement",
359         "manual_analysis",
360         "test_result",
361         "bcp07_kev_assertion",
362         "other"
363     ]
364 },
365 "source": {
366     "type": "string"
367 },
368 "url": {
369     "type": "string",
370     "format": "uri"
371 },
372 "tool": {
373     "$ref": "#/$defs/tool"
374 },
375 "reference": {
376     "$ref": "#/$defs/assessmentReference"
377 },
378 "details": {
379     "type": "object",
380     "additionalProperties": true
381 }
382 },
383 },
384 "tool": {
385     "type": "object",
386     "additionalProperties": false,
387     "required": [
388         "name"
389     ],
390     "properties": {
391         "name": {
392             "type": "string"
393         },
394         "version": {
395             "type": "string"
396         }
397     }
398 },
399 "assessmentReference": {
400     "type": "object",
401     "additionalProperties": false,
402     "properties": {
403         "uuid": {
404             "type": "string",
405             "format": "uuid"
406         },

```

```
407     "id": {
408         "type": "string"
409     },
410     "url": {
411         "type": "string",
412         "format": "uri"
413     }
414 },
415 "anyOf": [
416     {
417         "required": [
418             "uuid"
419         ]
420     },
421     {
422         "required": [
423             "id"
424         ]
425     },
426     {
427         "required": [
428             "url"
429         ]
430     }
431 ],
432 },
433 "reference": {
434     "type": "object",
435     "additionalProperties": false,
436     "required": [
437         "id",
438         "url"
439     ],
440     "properties": {
441         "id": {
442             "type": "string"
443         },
444         "url": {
445             "type": "string",
446             "format": "uri"
447         }
448     }
449 },
450 "gcveRoot": {
451     "type": "object",
452     "additionalProperties": false,
453     "description": "GCVE metadata describing the origin and identity
454                     of the NKEV assessment.",
455     "properties": {
456         "origin_uuid": {
457             "type": "string",
```

```
457         "format": "uuid"
458     },
459     "gna": {
460         "type": "integer",
461         "minimum": 0,
462         "maximum": 65535
463     },
464     "object_uuid": {
465         "type": "string",
466         "format": "uuid"
467     }
468 }
469 }
470 }
471 }
```


5 References

- GCVE BCP-07 defines separate but interoperable KEV assertion and NKEV assessment objects.
- `gcve-eu-kev` scripts - CISA KEV and ENISA CNW EUVD to GCVE BCP-07 Converter: <https://github.com/gcve-eu/gcve-eu-kev>

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